

Communication-Efficient Cooperative Highway Merging

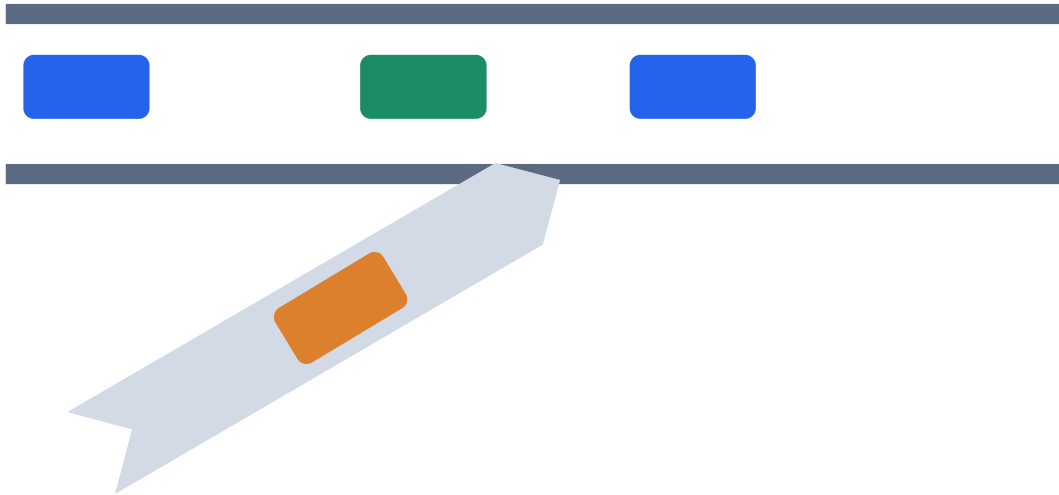
Vision-Based Connected Autonomous Vehicles

Student project – Mentored by:

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Core question: How much—and what type of information—should vehicles exchange to merge safely and efficiently?

The intuitive problem



- A ramp vehicle needs a safe gap.
- Main-lane vehicles may need to slow down, accelerate or change lane.
- Each vehicle sees only part of the scene.
- V2V communication can reveal information that another vehicle cannot see.
- But transmitting everything is expensive and may be unnecessary.

Research gap

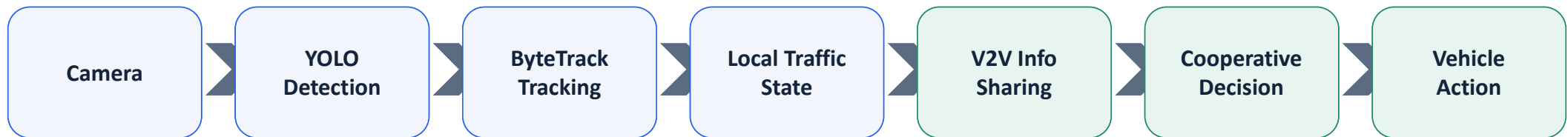
Move beyond “Can vehicles cooperate?” to “What information is actually needed for cooperation?”

- Cooperative highway merging using RL/MARL is already well studied.
- V2X-assisted merging and cooperative perception are also established.
- The stronger gap is the trade-off between information richness and communication cost.
- The project can test whether compact, task-relevant information is enough for safe merging.

Proposed RQ

How does the amount and type of shared perception information affect safety, efficiency and communication cost in cooperative highway merging?

System pipeline



- Computer vision creates the local observation; communication adds selected information from other CAVs; MARL turns the combined observation into cooperative actions.

What information should vehicles exchange?

1. Local only

Own camera perception | Communication load: None

2. State sharing

Position, speed, acceleration, lane, intent | Communication load: Very low

3. Semantic / object sharing

Detected objects: type, position, velocity, confidence | Communication load: Low

4. Compact feature sharing

Compressed intermediate perception features | Communication load: Medium

Research question

How much information is actually necessary for effective cooperative merging?

Semantic messages vs. compact features

Semantic / object sharing

Interpretable information after perception

Camera → Detector / Tracker → Semantic Message → V2V

Example message

Vehicle ID: 17
Position: 31.2 m ahead | Relative speed: -4.3 m/s
Lane: Main lane | Confidence: 0.93 | Intent: Yield

- Human-interpretable
- Small messages
- Potentially model-independent
- Interpretation happens before transmission

Compact feature sharing

Learned representation before final detection

Camera → Neural Encoder → Compact Features → V2V → Fusion

Example representation

ROI: merge zone
Feature vector: [0.82, -0.14, 0.63, ..., 0.41]
The receiver fuses this representation with its own features.

- Richer visual information
- Information shared before final detection
- Useful when perception is ambiguous / occluded
- Higher communication requirement

Scope assumption

Homogeneous or representation-compatible perception models

Main experiment: information richness vs. communication cost

A

Local only
Own camera
No communication

B

State sharing
Position + speed + lane + intent
Very low bandwidth

C

Semantic sharing
Detected / tracked objects
Low bandwidth

D

Compact features
Compressed neural features
Medium bandwidth

Same merging scenarios + same cooperative MARL framework + same perception task

Test conditions

Normal traffic | High density | Occlusion | Visual degradation

Evaluation

Safety: collisions, TTC, merge success, merge time, throughput
Communication: message size, data rate, latency

Hypothesis

Richer information may improve cooperative merging, but the improvement may not justify the additional communication cost.

What should we measure?

Driving safety

- Collision rate
- Time-to-collision
- Unsafe gaps

Merging performance

- Merge success
- Merge time
- Traffic throughput

Perception

- Detection accuracy
- Tracking quality
- Occlusion robustness

Communication

- Bytes/message
- Messages/second
- Total data
- Latency sensitivity

Recommended tool stack

Simulation: CARLA

Highway merging, cameras, weather, traffic

Vision: OpenCV + YOLO

Vehicle detection from RGB frames

Tracking: ByteTrack

Maintain vehicle identities across frames

MARL: PyTorch + PPO/IPPO/MAPPO

Learn cooperative merging decisions

Communication: Python V2V abstraction

Control exactly what agents exchange

Evaluation: Pandas + Matplotlib

Analyze safety, traffic and bandwidth metrics

Keep the project manageable

- Do not invent a new YOLO model, tracker and MARL algorithm at the same time.
- Use pretrained vision models and a standard MARL baseline.
- The scientific contribution is the controlled comparison of information-sharing strategies.
- Start with state sharing vs object sharing; feature sharing is an extension if time permits.
- Use perfect CARLA ground-truth state as an upper-bound reference.
- Only add packet loss/latency after the core experiment works.

Expected contribution

Key scientific idea

The best cooperative system may not be the one that shares the most information, but the one that shares the right information for the merging task.

- Target contribution: quantify the safety–efficiency–communication trade-off for vision-based cooperative highway merging.

Starting literature

- Chen et al. — "Deep multi-agent reinforcement learning for highway on-ramp merging in mixed traffic." *IEEE Transactions on Intelligent Transportation Systems* 24.11 (2023).
- Ibork et al. — "Z-Merge: Multi-Agent Reinforcement Learning for On-Ramp Merging with Zone-Specific V2X Traffic Information." *arXiv preprint arXiv:2511.14910* (2025).
- Parada et al. — "An end-to-end collaborative learning approach for connected autonomous vehicles in occluded scenarios." *2023 IEEE 26th International Conference on Intelligent Transportation Systems (ITSC)*. IEEE, 2023.
- Yuan et al. — "Keypoints-based deep feature fusion for cooperative vehicle detection of autonomous driving." *IEEE Robotics and Automation Letters* 7.2 (2022): 3054-3061..
- Ren et al. — "Collaborative joint perception and prediction for autonomous driving." *Sensors* 24.19 (2024): 6263.